

Math 252
Practice Questions

1. Find the explicit solution of the initial-value problem:

$$\frac{dP}{dt} = P - P^2, \quad P(0) = 2.$$

Answer: $P = \frac{2e^t}{2e^t - 1}.$

2. Solve the differential equation

$$x^3 \frac{dy}{dx} + 4x^2 y = \cos(3x).$$

Answer: $x^4 y = \frac{x}{3} \sin(3x) + \frac{1}{9} \cos(3x) + C.$

3. Find the solution of the initial-value problem

$$xy' + 2y = 3x, \quad y(1) = 0.$$

Answer: $y = x - 1/x^2$

4. Solve $\frac{dy}{dx} = \frac{y^2}{xy + x^2}.$

Answer: $y + x \ln |y| = cx.$

5. Solve the differential equation

$$y' = \frac{x + 3y}{3x + y}.$$

Answer: $\ln \left| \frac{1 + y/x}{(1 - y/x)^2} \right| = \ln |x| + C$

6. Solve the differential equation $xy' - y = 2y^3.$

Answer: $y^{-2} = cx^{-2} - 2.$

7. Solve the differential equation: $y' = y(xy^3 - 1).$ Give an explicit solution.

Answer: $y = \sqrt[3]{\frac{3}{3x + 1 + ce^{3x}}}.$

8. Solve the differential equation: $x \frac{dy}{dx} + y = \frac{e^{2x}}{xy^2}.$

Answer: $4y^3 x^3 = (6x - 3)e^{2x} + C.$

9. Find the value of the constant k so that the differential equation

$$(y^2 \cos x - 3x^2 y - 2x) dx + (ky \sin x - x^3 + 4y) dy = 0$$

is exact and then solve the equation.

Answer: $k = 2$ and the solution is

$$y^2 \sin x - x^3 y - x^2 + 2y^2 = C.$$

10. Solve $\frac{dy}{dx} = \frac{1 - x - y}{x + y}.$

Answer: $(x + y)^2 = 2x + c.$

11. Find the explicit solution of the initial-value problem: $y' = (4x + y)^2, \quad y(0) = 2.$

Answer: $y = 2 \tan(2x + \frac{\pi}{4}) - 4x.$

12. Solve the differential equation

$$x^2 y' = xy + y^4.$$

Answer: $\frac{1}{y^3} = \frac{-3}{2x} + \frac{C}{x^3}$

13. Solve the following differential equation:

$$(1 + xye^x) dx + (xe^x + x \cos y) dy = 0.$$

Answer: $\ln|x| + ye^x + \sin y = c.$

14. The rate of growth of a population of bacteria is proportional to the size of the population at all time. If 100 bacteria are present initially, and 500 are present after 20 minutes, how many will be present after 30 minutes?

Answer: 1118 bacteria

15. A tank contains 200 liters of fluid in which 100 grams of salt is dissolved initially. Brine containing 5 grams of salt per liter is then pumped into the tank at the rate 4 L/min; the well-mixed solution is pumped out at the same rate.

(a) Find the mass $m(t)$ of salt in the tank as a function of time t .

(b) Find $\lim_{t \rightarrow \infty} m(t)$.

(c) After how much time do we have 600 grams of salt in the tank?

Answers: (a) $m(t) = 1000 - 900e^{-t/50}$ (b) 1000 (c) 40.5 minutes.

16. According to Newton's Law of Cooling, the rate of change of temperature of a body is proportional to the difference in temperature between the body and the surrounding air, i.e.,

$$\frac{dT}{dt} = -k(T - T_a)$$

where T is the body temperature, T_a is the surrounding air temperature, and $k > 0$ is a constant.

(a) Solve the differential equation to find $T(t)$, assuming T_a is constant and $T(0) = T_0$.

(b) In a murder investigation, a corpse was found by Sherlock Holmes at exactly 8:00 pm. Being alert, he immediately measures the temperature of the body and finds it to be 25°C . Two hours later, Holmes again measures the temperature of the corpse and finds it to be 15°C . If the room temperature is 10°C , when did the murder occur? Assume that the temperature of the body at the time of the murder was 37°C .

Answers: (a) $T = T_a + (T_0 - T_a)e^{-kt}$, (b) at about 6:56 pm.

17. A differential equation describing the velocity
- v
- of a falling mass subject to air resistance proportional to instantaneous velocity is

$$m \frac{dv}{dt} = mg - kv$$

where $k > 0$ is a constant of proportionality.

(a) Solve the differential equation subject to the initial condition $v(0) = v_0$.

(b) Find $\lim_{t \rightarrow \infty} v(t)$. This limit is called the terminal velocity.

(c) A mass of 3 kg is falling from rest (i.e., $v_0 = 0$). After how much time will its velocity be 20 m/s? Use $g = 9.8 \text{ m/s}^2$ and $k = 0.5 \text{ N/(m/s)}$.

Answers: (a) $v = \frac{mg}{k} + (v_0 - \frac{mg}{k})e^{-kt/m}$ (b) mg/k (c) 2.5 seconds.

18. A tank in the form of a circular cylinder is leaking water through a small hole at the bottom. The height
- h
- of water in the tank is described by

$$\frac{dh}{dt} = -\frac{a}{A} \sqrt{2gh}$$

where A is the cross-section area of the tank and a is the area of the hole.

- (a) Solve the DE if the initial height of water is h_0 .
 (b) Suppose the tank is 3 m high with a radius of 40 cm and the hole is circular with a radius of 1.5 cm. How long will it take to empty the tank if it was initially full? Use $g = 9.8 \text{ m/s}^2$.

Answers: (a) $h = (\sqrt{h_0} - \frac{kt}{2})^2$ where $k = \frac{a}{A}\sqrt{2g}$ (b) 9.3 minutes

19. Find the general solution of the differential equation

$$xy'' - (1 + 3x)y' + 3y = 0$$

given that $y_1 = e^{3x}$ is a solution of the equation.

Answer: $y = c_1 e^{3x} + c_2(1 + 3x)$

20. Find the solution of the initial-value problem

$$y'' - 4xy' + (4x^2 - 2)y = 0, \quad y(0) = 1, \quad y'(0) = 2$$

given that $y_1 = e^{x^2}$ is a solution of the equation.

Answer: $y = e^{x^2} + 2xe^{x^2}$

21. Consider the differential equation

$$x^2 y'' - 3x y' + k y = 0, \quad (x > 0).$$

Find the general solution if (a) $k = -5$, (b) $k = 4$, (c) $k = 13$.

Answers:

- (a) $y = c_1 x^5 + c_2 x^{-1}$
 (b) $y = c_1 x^2 + c_2 x^2 \ln x$
 (c) $y = x^2(c_1 \cos(3 \ln x) + c_2 \sin(3 \ln x))$

22. Find the form of a particular solution y_p in the following. (Leave your answers with undetermined coefficients.)

- (a) $y'' + 4y = 5x + \sin(2x)$
 (b) $y'' - 2y' + y = 3x^2 + 5xe^x$
 (c) $y'' + 2y' + 5y = 3e^{-x} + 4e^{-x} \sin(2x) - 2 \cos(3x)$

Answers:

- (a) $y_p = Ax + B + Cx \cos(2x) + Dx \sin(2x)$
 (b) $y_p = Ax^2 + Bx + C + x^2(Dx + E)e^x$
 (c) $y_p = Ae^{-x} + xe^{-x} [B \cos(2x) + C \sin(2x)] + D \cos(3x) + E \sin(3x)$

23. Use the method of undetermined coefficients to find the general solution of

$$y'' + y' - 6y = 4e^{3x} + 5 \cos(2x).$$

Answer: $y = c_1 e^{-3x} + c_2 e^{2x} + \frac{2}{3} e^{3x} - \frac{25}{52} \cos(2x) + \frac{5}{52} \sin(2x)$

24. Use the method of undetermined coefficients to find the general solution of

$$y'' - 5y' + 6y = x + 2 + 3e^{2x}.$$

Answer: $y = c_1 e^{3x} + c_2 e^{2x} + \frac{x}{6} + \frac{17}{36} - 3xe^{2x}$

25. Solve

$$y'' + 4y = \sec(2x).$$

Answer: $y = c_1 \cos(2x) + c_2 \sin(2x) + \frac{1}{2} x \sin(2x) + \frac{1}{4} \cos(2x) \ln|\cos(2x)|$

26. Use the method of variation of parameters to find the general solution of

$$x^2 y'' - 2xy' + 2y = x^4 e^{-x}.$$

Simplify your answer.

Answer: $y = c_1 x^2 + c_2 x + x e^{-x}(x + 2)$

27. Find the solution of the initial-value problem

$$x^2 y'' - xy' + y = 2x, \quad y(1) = 2, \quad y'(1) = 3.$$

Answer: $y = 2x + x \ln x + x(\ln x)^2$

28. A mass of 3 kg is attached to a spring with constant 12 N/m (there is no damping). The mass is released from 20 cm below equilibrium with an upward velocity of 0.6 m/s.

- (a) Find the position of the mass as a function of time $x(t)$.
- (b) Find the period and amplitude of $x(t)$.
- (c) Find the first time $t > 0$ when the mass reaches its lowest point.

Answers: (a) $x = 0.2 \cos(2t) - 0.3 \sin(2t)$. (b) π seconds (c) 2.65 seconds

29. A mass of 2 kg is attached to a spring whose spring constant is 3 N/m. The medium offers a damping force of magnitude 5 times the instantaneous velocity.

- (a) Determine the steady-state position of the mass if it is driven by an external force (in Newton) of magnitude $F(t) = 4 \cos(2t) + 3 \sin(2t)$.
- (b) Express your answer in (a) in the form $x(t) = A \sin(\omega t + \phi)$.

Answers:

(a) $x_p(t) = -\frac{2}{5} \cos(2t) + \frac{1}{5} \sin(2t)$

(b) $x_p(t) = \frac{\sqrt{5}}{5} \sin(2t - 1.107)$

30. A 1 kg mass is attached to a spring with spring constant 2 N/m and a damping device with $\beta = 3$ N/(m/s).

- (a) Find the position $x(t)$ of the mass if it is released 50 cm below equilibrium with no initial velocity.
- (b) A force (in Newton) of the form $F(t) = F_0 \cos t$ is applied to the system. Find the positive value of F_0 so that the amplitude of the steady-state solution equals 1 m.

Answers:

(a) $x(t) = e^{-t} - 0.5e^{-2t}$

(b) $F_0 = \sqrt{10}$

31. A mass of 1 kg is attached to a spring whose spring constant is 5 N/m. The medium offers a damping force of magnitude 4 times the instantaneous velocity.

- (a) Find the position $x(t)$ of the mass if it is released 50 cm below equilibrium with an initial downward velocity of 1 m/s.
- (b) Find all times $t > 0$ when the mass passes through equilibrium position.

Answers:

(a) $x(t) = e^{-2t}(2 \sin t + \frac{1}{2} \cos t)$

(b) $t = n\pi - 0.245$, for all $n = 1, 2, 3, \dots$

32. A mass of 1 kg is attached to a spring whose spring constant is 6 N/m. The medium offers a damping force of magnitude 5 times the instantaneous velocity. An external force (in Newton) of the form $F_e = 3 \sin(2t) - \cos(2t)$ is applied to the system. Find the position of the mass $x(t)$ if it is released 50 cm above equilibrium with a downward velocity of 1 m/s.

Answer: $x(t) = \frac{1}{2}e^{-2t} - \frac{9}{13}e^{-3t} - \frac{4}{13} \cos(2t) - \frac{1}{26} \sin(2t)$

33. Find the first five nonzero terms of a power series solution of the initial-value problem

$$y'' + x^2 y' - 3y = 0, \quad y(0) = 3, \quad y'(0) = -2.$$

For which values of x does the series converge?

Answer: $y = 3 - 2x + \frac{9}{2}x^2 - x^3 + \frac{31}{24}x^4 + \cdots$. The series converges for all x .

34. Find the first five nonzero terms of a power series solution of the initial-value problem

$$(x+1)y'' - (2-x)y' + y = 0, \quad y(0) = 2, \quad y'(0) = -1.$$

For which values of x does the series converge?

Answer: $y = 2 - x - 2x^2 - \frac{1}{3}x^3 + \frac{1}{2}x^4 - \frac{1}{30}x^5 + \cdots$. The series converges at least for all $-1 < x < 1$.

35. Use power series to solve the differential equation: $y'' - (1+x)y = 0$.

Answer: $y = c_1(1 + \frac{x^2}{2} + \frac{x^3}{6} + \cdots) + c_2(x + \frac{x^3}{6} + \frac{x^4}{12} + \cdots)$

36. Use power series to solve the differential equation: $y'' + e^x y = 0$.

Answer: $y = c_1(1 - \frac{x^2}{2} - \frac{x^3}{6} + \cdots) + c_2(x - \frac{x^3}{6} - \frac{x^4}{12} + \cdots)$

37. Evaluate the following.

(a) $\mathcal{L}\{f(t)\}$ for

$$f(t) = \begin{cases} 1, & 0 \leq t < 2 \\ 2, & 2 \leq t < 4 \\ -1, & t \geq 4 \end{cases}$$

(b) $\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s+3}\right\}$

(c) $\mathcal{L}\{te^{-2t}\sin(3t)\}$

(d) $\mathcal{L}^{-1}\left\{\frac{s+4}{s^2+2s+5}\right\}$

(e) $\mathcal{L}\{\cos t \mathcal{U}(t-\pi)\}$

Answers: (a) $\frac{1}{s} + \frac{e^{-2s}}{s} - \frac{3e^{-4s}}{s}$ (b) $e^{-3(t-2)}\mathcal{U}(t-2)$ (c) $\frac{6(s+2)}{((s+2)^2+9)^2}$

(d) $e^{-t}\cos(2t) + \frac{3}{2}e^{-t}\sin(2t)$ (e) $\frac{-se^{-\pi s}}{s^2+1}$

38. Use Laplace transform to solve the initial-value problem

$$y''(t) + 4y'(t) + 3y(t) = 3e^{-2t}, \quad y(0) = 2, \quad y'(0) = 3.$$

Answer: $y(t) = 6e^{-t} - e^{-3t} - 3e^{-2t}$.

39. Use Laplace transform to solve the initial-value problem

$$y'(t) + 3y(t) = \sin(2t), \quad y(0) = 4.$$

Answer: $y(t) = \frac{54}{13}e^{-3t} - \frac{2}{13}\cos(2t) + \frac{3}{13}\sin(2t)$.

40. Use Laplace transform to find the solution of the initial-value problem:

$$y''(t) + 9y(t) = e^{-t} + \cos(3t), \quad y(0) = 0, \quad y'(0) = 1.$$

Simplify your answer.

Answer: $y(t) = \frac{1}{10}e^{-t} - \frac{1}{10}\cos(3t) + \frac{11}{30}\sin(3t) + \frac{1}{6}t\sin(3t)$

41. Use Laplace transform to solve the initial-value problem

$$y''(t) + 2y'(t) + 5y(t) = 3e^{-t} \cos(2t), \quad y(0) = 1, \quad y'(0) = 2.$$

Answer: $y(t) = e^{-t} \cos(2t) + \frac{3}{2}e^{-t} \sin(2t) + \frac{3}{4}te^{-t} \sin(2t)$

42. Use Laplace transform to find the solution of the following initial value problem.

$$y'(t) + 5y(t) = 3 + \delta(t - 2), \quad y(0) = 4.$$

Answer: $y(t) = \frac{17}{5}e^{-5t} + \frac{3}{5} + e^{-5(t-2)}\mathcal{U}(t - 2)$

43. Solve the initial value problem

$$y'(t) + 2y(t) = f(t), \quad y(0) = 0$$

where

$$f(t) = \begin{cases} 1, & 0 \leq t < 1, \\ 2, & 1 \leq t < 2, \\ 0, & t \geq 2. \end{cases}$$

Compute the value of $y(t)$ when $t = 1.5$.

Answer:

$$y(t) = \left(\frac{1}{2} - \frac{1}{2}e^{-2t}\right) + \left(\frac{1}{2} - \frac{1}{2}e^{-2(t-1)}\right)\mathcal{U}(t - 1) - \left(1 - e^{-2(t-2)}\right)\mathcal{U}(t - 2)$$

and $y(1.5) = 0.791$

44. The differential equation for the current $i(t)$ in an LR series circuit is

$$L \frac{di}{dt} + Ri(t) = V(t).$$

If the inductance and resistance are respectively $L = 1 \text{ H}$, $R = 10 \Omega$ and

$$V(t) = \begin{cases} 1, & 0 \leq t < 2 \\ 0, & t \geq 2. \end{cases}$$

Find the current $i(t)$ if $i(0) = 0$.

Answer:

$$i(t) = \begin{cases} \frac{1}{10}(1 - e^{-10t}), & 0 \leq t < 2 \\ \frac{1}{10}(e^{-10(t-2)} - e^{-10t}), & t \geq 2 \end{cases}$$

45. Solve the initial-value problem

$$y''(t) + y(t) = 2e^{-t} + \delta(t - \frac{\pi}{2}) + \delta(t - \pi), \quad y(0) = 2, \quad y'(0) = 1.$$

Write your answer as a piecewise-defined function.

Answer:

$$y(t) = \begin{cases} \cos t + 2 \sin t + e^{-t}, & 0 \leq t < \frac{\pi}{2} \\ 2 \sin t + e^{-t}, & \frac{\pi}{2} \leq t < \pi \\ \sin t + e^{-t}, & t \geq \pi \end{cases}$$

46. Evaluate $\mathcal{L}^{-1} \left\{ \frac{1}{(s-3)(s+4)} \right\}$ by using convolution.

Answer: $\frac{1}{7}e^{3t} - \frac{1}{7}e^{-4t}$

47. Evaluate $\mathcal{L}^{-1} \left\{ \frac{1}{(s^2+1)^2} \right\}$ by using convolution.

Answer: $\frac{\sin t - t \cos t}{2}$

48. Find the general solution of $\mathbf{X}'(t) = A\mathbf{X}(t)$ if

(a) $A = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$.

(b) $A = \begin{pmatrix} -1 & -4 \\ 1 & -1 \end{pmatrix}$

(c) $A = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$

Answers:

(a) $\mathbf{X}(t) = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{5t} + c_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-t}$

(b) $\mathbf{X}(t) = \left[c_1 \begin{pmatrix} -2 \sin 2t \\ \cos 2t \end{pmatrix} + c_2 \begin{pmatrix} 2 \cos 2t \\ \sin 2t \end{pmatrix} \right] e^{-t}$

(c) $\mathbf{X}(t) = c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^t + c_2 \left[\begin{pmatrix} 2 \\ 1 \end{pmatrix} t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right] e^t$

49. Solve the initial value problem.

$$\mathbf{X}'(t) = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix} \mathbf{X}(t), \quad \mathbf{X}(0) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Answer:

$$\mathbf{X}(t) = \frac{5}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{4t} - \frac{1}{2} \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{2t} = \begin{pmatrix} \frac{5}{2}e^{4t} - \frac{1}{2}e^{2t} \\ \frac{5}{2}e^{4t} - \frac{3}{2}e^{2t} \end{pmatrix}$$

50. Use the method of variation of parameters to find the general solution of the system

$$\begin{cases} \frac{dx}{dt} = 2y + e^t \\ \frac{dy}{dt} = 3y - x - e^t \end{cases}$$

Answer:

$$\begin{cases} x = 2c_1 e^t + c_2 e^{2t} + (4t + 3)e^t \\ y = c_1 e^t + c_2 e^{2t} + (2t + 3)e^t \end{cases}$$

51. Solve the initial value problem

$$\mathbf{X}'(t) = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} \mathbf{X}(t) + \begin{pmatrix} 2 \\ t \end{pmatrix}, \quad \mathbf{X}(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Answer: $\mathbf{X}(t) = \begin{pmatrix} 2 \\ 2 \end{pmatrix} e^t + \begin{pmatrix} 2 \\ 6 \end{pmatrix} e^{-t} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} t - \begin{pmatrix} 4 \\ 7 \end{pmatrix} = \begin{pmatrix} 2e^t + 2e^{-t} + t - 4 \\ 2e^t + 6e^{-t} + 2t - 7 \end{pmatrix}$